

MS in Management Science & Engineering Advising Town Hall

Fall 2025

Agenda

1. Welcome & Introductions
2. Bootcamps
3. Waiver Exams
4. Registration Guidelines
5. Degree Requirements
6. Program Requirements
7. Fall Electives
8. CBS Courses / Cross Registration/ Course Approvals
9. Fieldwork
10. Upcoming Events

Academic Team



CHRIS LEE @ CL4125
*Academic & Student Affairs
Assistant Director*



CHRISTINE CHAN @ CC5201
*Academic & Student Affairs
Senior Associate Director*



WINSOR YANG @ WY2396
*Academic & Student Affairs
Assistant Director*



SABRINA LI @ SL5163
*Academic & Student Affairs
Coordinator*



CINDY BORGEN @ CB3871
*Academic & Student Affairs
Coordinator*



ABHISHANK KHADKA @ AK5211
*Academic & Student Affairs
Operations and Data Analyst*



KELLY KATSIGRIS @ KK3813
*Academic & Student Affairs
Advisor*

Academic Office Hours

- Summer Group Office Hours- Open Q&A
 - July 15 @ 3 PM EST:
[Zoom Link](#)
- Fall '25 Office Hours
Monday-Thursday 3-4pm
- We're a resource for you! We can support you with:
 - Course Registration
 - Tracking degree progress
 - Starting your CPT/OPT
 - Connect you to campus resources and opportunities

...& so much more!

Career Team



LUCY MAHBUB @ LAM2308
Director, Career Placement



CINDY MEJIA @ CM4328
*Career Placement Officer for
MSOR/IE*



DAVID FITZGERALD @ DF2774
*Career Placement Officer for
MSFE*



SARA YLLESCAS @ SMY2124
*Career Placement Officer for
MSBA*



KARINA IKEDA @ KK3809
*Career Placement Officer for
MSE*

Program Director



Prof. Hardeep Johar

Online Summer Bootcamps

- DROM B8000 Optimization and Simulation Bootcamp - *Required*
 - Self-paced online modules hosted on Courseworks (beginning early August)
 - Pass grade
- IEOR E4399 MSE Quantitative Bootcamp - *Required*
 - Self-paced online modules hosted on Courseworks (beginning early August)
 - Module quizzes to check understanding
 - Pass grade
- IEOR E4502 Python for Analytics Bootcamp - *Required*
 - In-person modules
 - 4-day in-person class taught by Instructors (8/19, 8/20, 8/21, 8/23)
 - 4-day in-person tutorials facilitated by Course Assistants (8/19, 8/20, 8/21, 8/23)
 - Final proficiency exam (8/29th @2:00pm EST)

Proficiency & Waiver Exams

(Deadline to Register - August 20th)

Probability, Part 1 Waiver Exam

Thursday, Aug 28th @ 9am (in-person ONLY)

- If Pass: IEOE 4101 (3.0 credits) → IEOE 4100 (1.5 credits)*
- If Not Pass: IEOE 4101 (3.0 credits) → remains the same

Python for Analytics - Waiver Exam

Friday, Aug 29th @2:00pm EST

- If Pass: Can waive IEOE E4501 Tools for Analytics requisite
- If Not Pass: IEOE E4501 Tools for Analytics co-requisite / pre-requisite not waived.

***Students may only take each exam once during their studies.**

Overview of Degree Requirements

1. No Pass/Fail grades, No R grade in credit bearing courses
2. Must achieve at least a B- (2.67) in Core courses and overall GPA
3. All courses must be completed at Columbia.
4. ENGI E4000 pass grade is a requirement
5. Satisfy (18) credits minimum from IEOR and your Program requirements. Course codes include: IEOR, CSOR, ORCS and IEME
6. Good academic standing; uphold academic and professional integrity

Course Modality & Registration

All classes are expected to be taught in-person, on-campus.

- Fall 2025 Vergil [Course Registration](#):
 - Registration begins July 14th at 9:30AM EST
 - Registration (again) Aug. 4 at 9:30AM EST
 - Add/Drop Period: Sept. 2 - 12
 - CBS Add/Drop Period Aug 6 - Sept 5
- Last day to drop a course without financial penalty is [Sept. 12th, 2025](#).
 - Includes waitlists
- Registration Credit Limit (includes waitlists):
 - 1st & 2nd Semester - 15 credits
 - 3rd Semester (Fall '26) - 12 credits
 - If you would like to go above these credits, please meet with a member of our team.
- Admission deposits are credited to your student account *after* Sept. 13th
- For course descriptions, refer to the [Directory of Classes](#)
- For other important deadlines/dates, refer to the [Academic Calendar](#)

CORE REQUIREMENTS FOR MS PROGRAMS

MS&E Core

- IEOR E4004.001 Optimization Models (F25)
- IEOR E4101.001 Prob, Stats & Simulations (F25)
- IEOR E4111 Operations Consulting (year long)
- IEOR E4102 Stochastic Models for MS&E (Sp26)
- IEOR E4399 MSE Quantitative Bootcamp (Su25)
- DROM B8000 Opt & Simulation Bootcamp (Su25)
- ENGI E4000 PDL (F25 & Sp26)

Semi-Core

Decision, Risk and Operations	9 credits	
Analysis	≥ 3 credits	9 credits total
Management	≥ 3 credits	

At least 6 courses (18 credits) must be in the IEOR department (IEOR, IEME, CSOR designation).

mse.ieor.columbia.edu/content/curriculum

MS&E (12 Credits) - Fall '25

Register for:

1. IEOR E4004 Section 003 Optimization Models
2. IEOR E4101 Section 003 Probability, Statistics & Simulation
3. IEOR E4111 Operations Consulting (Year long course)
4. **Elective (3 credits)**

The Department will register you into the below courses:

- IEOR E4399 MSE Quantitative Bootcamp (0pt)
- DROM B8000 Optimization & Simulation Bootcamp (0pt)
- ENGI E4000 Professional Development & Leadership (0pt)

IEOR E4111 Operations Consulting

Mark your calendars!

First Class Session:

- Thursday, August 28
- Lunch @12PM (Location TBD)
- Lecture @2 PM (Location TBD)
- Business attire
- **Attendance is mandatory**

Review course structure, form consulting teams,
discover available topics, and bid on desired projects



ELECTIVES FOR MS PROGRAMS

Interest in Analytics

Course Number	Course Title	Comments
IEORE4404	Simulation	MSE Breadth; Priority to MSOR (Core)
IEORE4418	Transportation Analytics & Logistics	MSE Analysis Semi-Core
IEORE4501	Tools for Analytics	MSE Analysis Semi-Core
IEORE4506	Designing Digital Operating Models	MSE Management Semi-Core
IEORE4507	Healthcare Operations Management	MSE Analysis Semi-Core
IEORE4520	Applied Systems Engineering	MSE Analysis Semi-Core
IEORE4523	Data Analytics	MSE Analysis Semi-core; Priority to MSBA; IEOR4501/waiver required
IEORE4525	Machine Learning FE & OR	MSE Analysis Semi-Core
IEORE4526	Analytics on the Cloud	MSE Analysis Semi-Core; Prerequisites: IEOR4501/waiver & IEOR4523
IEORE4532	Visualization & Storytelling with Data	MSE Analysis Semi-Core, 1.5 credits; Term B, Meets all day on Fridays
IEORE4533	Performance, Objectives & Results Using Data Analytics	MSE Analysis Semi-Core; 1.5 credits; Term A, 3-day Blockweekend (9/12-9/14)
IEORE4534	Applied Analytics: from Data to Decisions	MSE Analysis Semi-Core
IEORE4540	Data Mining for Engineers	MSE Analysis Semi-Core
IEORE4572	Data Science Applications in Insurance and Banking	MSE Analysis Semi-Core
IEORE4573	Business Applications of Large Language Models (LLMs)	MSE Analysis Semi-Core, 1.5 credits; Term A
IEORE4574	AI Engineering Applications	MSE Analysis Semi-Core, 1.5 credits; Term B
IEORE4577	Applied Risk Analytics	MSE Analysis Semi-Core, 1.5 credits, Term B
IEORE4650	Business Analytics	MSE Analysis or DRO; Taught by Adam Elmachtoub
IEORE4742	Deep Learning For OR & FE	MSE Analysis Semi-Core; (Dates TBD)
ORCSE4529	Reinforcement Learning	MSE Analysis Semi-Core



Interest in Consulting

Course Number	Course Title	Comments
IEORE4506	Designing Digital Operating Models	MSE Management Semi-Core
IEORE4532	Visualization & Storytelling with Data	1.5 credits; Meets all day 10/18, 10/25, 11/1 9am to 5pm
IEORE4533	Performance, Objectives & Results Using Data Analytics	MSE Analysis Semi-Core; 1.5 credits; Meets all day Sep. 13th, 14th, 15th

Interest in Entrepreneurship

Course Number	Course Title	Comments
IEORE4200	Human-Centered Design & Innovation	MSE Management Semi-Core
IEORE4207	Human Factors: Performance	MSE Breadth

Interest in Financial Engineering

Course Number	Course Title	Comments
IEORE4402	Corporate Finance, Accounting & Investment Banking	MSE Management Semi-Core; May not take both IEOR4402 & 4403
IEORE4500	Applications Programming for FE	MSE Breadth; Priority to MSFE
IEORE4525	Machine Learning FE & OR	MSE Analysis Semi-Core
IEORE4540	Data Mining for Engineers	MSE Analysis Semi-Core
IEORE4576	Data-driven Method Finance	MSE Breadth
IEORE4602	Quantitative Risk Management	MSE Breadth; Priority to MSFE
IEORE4620	Pricing Models For Financial Engineering	MSE Breadth
IEORE4700	Intro to Financial Engineering	MSE Breadth
IEORE4711	Global Capital Markets	MSE Management Semi-Core; May not take both IEOR4711 & FINC B8306
IEORE4723	FE for Environmental, Social & Corporate Gov. Finance	MSE Breadth; 1.5 credits; A-Term (First half of semester)
IEORE4724	Term Structures & Credit Models	MSE Breadth; Priority to MSFE
IEORE4728	Alternative Investments	MSE Breadth; 1.5 credits (Half term course)
IEORE4734	Foreign Exchange / Related Derivatives	MSE Breadth; 1.5 credits; B-Term (Second half of semester)
IEORE4735	Structured & Hybrid Products	MSE Breadth; Priority to MSFE
IEORE4741	Programing for Financial Engineering	MSE Breadth
IEORE4742	Deep Learning For OR & FE	MSE Analysis Semi-Core
IEORE4745	Applied Financial Risk Management	MSE Breadth

MSE Program Plan

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IN THE CITY OF NEW YORK

DEPARTMENT OF INDUSTRIAL ENGINEERING AND OPERATIONS RESEARCH

NAME: _____

NAME:

UNI:

PID:

CELL PHONE:

ADVISOR:

TERM	COURSE NUMBER	POINT S	COMPLETED	ADVISOR'S NOTES	DATE
Spring Fall 2015	Sample IEOR E4004	Sample 3.0	☐	Sample Approved Fall 2015 schedule	Sample 3/31/2014
	IEOR E4004 (core)	3.0	☐		
	IEOR E4111 (core)	3.0	☐		
	IEOR E4101 (core)	3.0	☐		
	IEOR E4102 (core)	3.0	☐		
DECISION, RISK & OPERATIONS (DRO) ELECTIVES (9 credits total)					
			☐		
			☐		
			☐		
ANALYSIS & MANAGEMENT ELECTIVES (9 credits total)					
ANALYSIS ELECTIVES (minimum 1 course, need 9 credits total for Analysis & Management Electives):					
			☐		

[Link to Program Plan](#)

Please use the above link to help plan your
MSE Program

Concentrations

Should you like to specialize in certain area, you have the option to follow a concentration.

MSE Concentrations:

- Operations & Analytics
- Pricing & Revenue Management
- Entrepreneurship & Innovation
- Logistics & Supply Chain Management
- Financial Technology (FinTech)
- Healthcare Management
- Product Management

Concentrations are *optional*.

Doctoral Level IEOR Courses

Course Number	Course Title	Notes
IEORE6613	Optimization I	4.5 credits
IEORE6711	Stochastic Modelling I	4.5 credits
IEORE6617	Machine Learning & High-Dimensional Data Analysis in OR	
IEORE8100	Topics Courses (TBA)	

All doctoral level IEOR courses require faculty approval

Business School Registration

- [Pre-approved CBS course list](#) - check this list to see if a CBS course is approved for your program. Courses not on this list require approval.
- If a CBS course is pre-approved for your program, **it does not** guarantee you a seat in the course.
- Additional questions regarding CBS courses? Reach out to the CBS team at: coursematch@gsb.columbia.edu (Beginning Fall 2025)
- [CBS Course Catalogue](#)
- Reminder: CBS Course Add/Drop period starts August 6th

Course Approvals

- Courses not on the pre-approved list for your degree program require approval from the academic team prior to taking the course (**not after the course is completed**).
- You can refer to the historic course approval list [here](#).
- If you do not see the course on the list, you must request approval from the academic team via the [course approval request form](#).
- **Reminder:** If you are considering taking any course outside of the IEOR department that is not pre-approved, you must request approval
- **Reminder:** Courses from SPS / QMSS classes are not approved
- **Reminder:** Programming courses from CBS are not approved

Cross Registration Outside of IEOR

IEOR students may “cross-register” for courses in other departments such as Statistics, Computer Science, Mathematics, Civil Engineering, or CBS:

- **Availability:** the list of classes available for cross-registration is determined by the host department, not IEOR. Please reach out to the corresponding office with questions about specific classes, timeline, and more.
- **Vergil:** If you cannot add a non-IEOR class on Vergil, it is likely unavailable to you for cross-registration.
- **Course Approval:** you must receive faculty approval for a cross-registered course to count towards your degree. Certain CBS courses are already pre-approved.

Fieldwork/CPT

Earn academic credit for your internship via IEOR 4999 - Fieldwork and CPT

Information & Application for Fieldwork/CPT can be found here:

<https://ieor.columbia.edu/masters-fieldwork>

- Must register for **IEOR 4999 Fieldwork** (1, 1.5, or 2 credits)
 - 1 / 1.5 during Fall and Spring, **2 units Summer only**
 - Maximum of **3 credits total** over program duration
 - Up to 20 hrs/week during Spring and Fall, full time (20+hrs/week) during Summer
- CPT is only available to F-1 Visa students after they have completed 2 consecutive semesters of full-time credits (12+). This means that for most students, Summer 2026 would be the first term you would engage in CPT as a F-1 Visa student.

Upcoming IEOR Events

- Department Kick-Off : Monday August 18th (required)
- SEAS Welcome Day : Thursday August 21 (required)
- SEAS PDL Day : Friday August 22 (required)
- Networking Dinners : TBA
- Faculty Showcase : Friday September 5th
- 1st & 2nd Year Social: TBA
- & more!



MSBA Career Placement Officer



KARINA IKEDA @ KK3806
Career Placement Officer for MSE

career@ieor.columbia.edu

Questions?

✉ Email us

Academic info@ieor.columbia.edu

Career Placement career@ieor.columbia.edu



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